

Jongseo Lee

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Research Interests

- Interpretable video understanding (concept bottleneck models)
- Diagnosing and improving Video-LLMs (motion and physical reasoning)
- Video world models and intuitive physics
- Video representation learning (multimodal, continual)

Education

M.S. in Computer Science and Engineering, Kyung Hee University, South Korea

Sep. 2023 – Aug. 2025

Advisor: Prof. Jinwoo Choi (Vision and Learning Lab)

Degree: Master of Engineering | GPA: 4.21 / 4.3

B.S. in Biomedical Engineering & Electronic Engineering, Kyung Hee University, South Korea

Mar. 2017 – Aug. 2023

Double major | Degree: Bachelor of Engineering | GPA: 3.95 / 4.3

Research Experience

Post-Master Researcher, Vision and Learning Lab, Kyung Hee University, South Korea

Sep. 2025 – Present

Leading research on video-LLM evaluation/diagnosis and explainable video understanding; international collaborations with UC San Diego (Prof. Tsui-Wei Weng, TRACE) and Princeton University (Jihoon Chung, DeltaDirect).

Graduate Research Assistant, Vision and Learning Lab, Kyung Hee University, South Korea

Sep. 2023 – Aug. 2025

Researched video class-incremental learning and explainable action recognition; collaborated with Intel Labs (Kyle Min) on ESSENTIAL; worked on ETRI-funded research projects.

Undergraduate Research Intern, Vision and Learning Lab, Kyung Hee University, South Korea

Feb. 2022 – Jul. 2023

Conducted research on video action recognition; co-led CAST (NeurIPS 2023) as a co-first author.

Publications

(* equal contribution, † corresponding author)

International Conferences & Journals

1. **Which Way Did It Move? Diagnosing and Overcoming Directional Motion Blindness in Video-LLMs**

Jongseo Lee*, Hyuntak Lee*, Sunghun Kim, Sooa Kim, Jihoon Chung, Jinwoo Choi†

NeurIPS 2026 [paper] [project] [code]

Also presented at the 2nd Workshop on Compositional Learning: Safety, Interpretability, and Agents (CompLearn), ICML 2026 (non-archival poster)

2. **TRACE: Trajectory-Aware Concepts for Explainable Video Understanding**

Wooil Lee*, Junpyo Hong*, Jongseo Lee, Tuomas Oikarinen, Tsui-Wei Weng, Jinwoo Choi†

NeurIPS 2026

3. **Disentangled Concepts Speak Louder Than Words: Explainable Video Action Recognition**

Jongseo Lee, Wooil Lee, Gyeong-Moon Park†, Seong Tae Kim†, Jinwoo Choi†

NeurIPS 2025, Spotlight (top 3.5%) [paper] [code]

4. **ESSENTIAL: Episodic and Semantic Memory Integration for Video Class-Incremental Learning**

Jongseo Lee*, Kyungho Bae*, Kyle Min, Gyeong-Moon Park†, Jinwoo Choi†

ICCV 2025, Highlight Presentation (top 2.9%) [paper] [code]

5. **CA²ST: Cross-Attention in Audio, Space, and Time for Holistic Video Recognition**

Jongseo Lee*, Joohyun Chang*, Dongho Lee, Jinwoo Choi†

IEEE TPAMI, 2025 [paper]

6. **PCBEAR: Pose Concept Bottleneck for Explainable Action Recognition**

Jongseo Lee, Wooil Lee, Gyeong-Moon Park†, Seong Tae Kim†, Jinwoo Choi†

CVPR Workshop on XAI4CV 2025, Spotlight (top 16.7%) [paper]

7. CAST: Cross-Attention in Space and Time for Video Action Recognition

Dongho Lee*, Jongseo Lee*, Jinwoo Choi†

NeurIPS 2023 [paper] [poster] [code]

Preprints

8. PCEvE: Part Contribution Evaluation Based Model Explanation for Human Figure Drawing Assessment and Beyond

Jongseo Lee*, Geo Ahn*, Seong Tae Kim†, Jinwoo Choi†

arXiv preprint, 2024 [paper]

Non-Archival Workshop Presentations

• BiAdA: Holistic Video Recognition via Bidirectional Adapter Fusion

Jongseo Lee*, Joohyun Chang*, Dongho Lee, Jinwoo Choi†

ICCV 2025 Workshop on Long Video Foundations (LongVidFoundations), non-archival poster

Domestic Publications

• Metaverse Interface with Haptic and Rigid Sense Feedback at a Low Cost

Jongseo Lee, Su Hyeon Kim, Sun Woong Jang, Jun Yeong Moon, Doug Young Suh†

Journal of Appropriate Technology, 8(2), 2022 [paper]

• KIISE 2024: Video Concept Bottleneck Model

• KIISE 2024: Efficient Video Class Incremental Learning via Class Token Re-Learning

• KIISE 2024: Audio-Video Cross Attention for Effective Video Action Recognition

Research Projects

Skeleton-Based Choreography Recognition for K-POP Dance Videos (SkeletonAR)

2026

Industry-funded research project with MetaRocket — skeleton-based recognition of fast, multi-person choreography; KOLAS-accredited third-party performance test in progress

Motivational Context Analysis of Interaction-Based Reactions in Video

2025

Funded research project with ETRI

Classification and Explanation Models for ASD Screening Based on Drawing Video Understanding

2024

Funded research project with ETRI — related publication: PCEvE

Deep Learning Models and Explainable AI for ASD Screening Drawing Data

2023

Funded research project with ETRI

Rule-Based Group Activity Recognition Video Dataset (AI Hub)

2022

National dataset construction project in collaboration with GAZZILABS — developed and validated AI recognition algorithms

Honors & Awards

• Spotlight Presentation, NeurIPS 2025 (top 3.5% of accepted papers)

2025

• Highlight Presentation, ICCV 2025 (top 2.9% of accepted papers)

2025

• Spotlight, CVPR Workshop on Explainable AI for Computer Vision

2025

Technical Skills

Programming: Python, C/C++ **Frameworks:** PyTorch

Systems: Large-scale video/multimodal model training and evaluation

Academic Service

Reviewer:

• NeurIPS 2026

• AAAI 2025

• CVPR Workshop on Explainable AI for Computer Vision (XAI4CV) 2026

• 2nd Workshop on Compositional Learning: Safety, Interpretability, and Agents (CompLearn), ICML 2026